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Multimodal Freight Transportation Network Design Problem for Reduction of Greenhouse Gas Emissions

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At the strategic level, multimodal freight network design problems are limited in that they reflect the costs and constraints associated with emissions. The network design problem (NDP) addressed in this study determines investment alternatives for minimizing total system cost, including costs related to greenhouse gas (GHG) emissions, while satisfying the emissions constraint. The NDP can accommodate improvement alternatives in transfer terminals as well as in road and railroad links. In this study, mode-specific travel time functions were used to represent the differences between the modes explicitly, even for the transfer facility. Emission factors were calculated by link travel speed and level of facilities. The empirical application to a container freight network from the Port of Pusan in southeastern South Korea showed the optimal investment strategy for meeting emissions reduction policy objectives. The conclusion is that investment should concentrate on railroad and terminal facilities to induce modal shift. In contrast, investment in congested roads might be a better option for corridors in which the level of railroad service is low and short-distance freight demands are dominant. These results indicate that the proposed network design problem can provide the appropriate investment strategy for reducing GHG emissions and therefore can be a useful approach with regard to GHG emissions reduction policy.

As global warming progresses, many countries have established greenhouse gas (GHG) emissions reduction targets (*1–3*), and emissions reduction is now a major issue for the transportation sector. Transportation contributes approximately 24% of total GHG emissions, and most of these emissions are produced by vehicles on the road (*4*). In South Korea, road transportation accounts for more than 90% of total transportation GHG emissions, and about 40% of these emissions come from the freight transportation sector (*5*). To achieve the emissions reduction target, freight transportation will be required to undertake challenging actions, since it relies extensively on road transportation, which is an energy-inefficient and high-emissions mode.

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On the basis of work of Dalkmann and Brannigan, GHG emissions reduction strategies are commonly classified into three categories: avoid, shift, and improve (*6*). Of those strategies, shift, which means promotion of the use of lower-emission modes, is particularly effective for freight, because the modes of freight transportation have large differences in emissions (*7*). Since the low competitiveness of rail transportation facilities is one of the main reasons for the current dependence on road transportation, enhancing the quality of facilities is required for modal shift. Litman (*8*) also mentioned that improving the quality of service provided by more resource-efficient modes and shifting freight to these modes would provide multiple benefits.

Of the various options for facility improvement, this study deals with a strategic network planning approach to determine long-term investments in transportation facilities. Strategic network planning has placed less emphasis on investing in energy-efficient modes. Although many countries have tried to increase their investment in low-emission modes in recent years, transportation network planning methods have not provided theoretical support. Therefore, considerations with regard to emissions reduction must be introduced into investment plans to meet emissions reduction goals.

The conventional network design problem (NDP) can be used as a strategic network planning approach. The NDP deals with expansions or additions of facilities in a transportation network by using a mathematical programming technique. The NDP can support the planner's decisions concerning investment with consideration of the users' responses to changes in the system within a bilevel framework. In addition, the NDP can provide the best combination of investment actions for the entire network by identifying the interactions between alternatives. This will allow analysis of alternatives for enhancing integration between modes or facilities by including various types of modes or facilities in a network. Previous studies provided a general framework of the NDP as well as models and solution techniques (*9–11*).

Few strategic NDP studies have dealt with a freight transportation network, and most of them did not consider GHG emissions reduction. Strategic freight NDP studies have mainly considered minimizing transportation cost and travel time (*12, 13*). One study considered environmental efficiencies according to the modes, but it did not explicitly consider GHG emissions and emission targets (*14*). Although a few studies have considered the minimization of GHG emissions as an objective, they did not deal with facility investment alternatives in freight networks. For example, a study by Kim et al. dealt with a freight transport mode-chain decision (*15*), and an NDP study by Dimitriou and Stathopoulos addressed passenger transportation (*16*). In contrast to strategic planning, many NDP studies for

service network design at the tactical level have dealt with GHG emissions reduction and constraints (*17–20*).

This study aims to establish an interregional freight NDP to support a GHG emissions reduction policy. At the strategic planning level, it determines the optimal combination of long-term investment alternatives in a discrete, bilevel NDP framework. For an interregional freight network, discrete decision variables are appropriate, because there are various improvement alternatives that change speed and capacity (*12*). This NDP deals with alternatives for improving or expanding existing facilities containing road and railroad links and transfer terminals without consideration of new facilities. To achieve the purpose of this study, the model minimizes the social costs due to GHG emissions along with other costs and contains the maximum emission as a constraint.

Since a discrete NDP has computational difficulties, many researchers have applied various solution methods to reduce or avoid the complexity, as mentioned in the study by Gao et al. (*11*). Recently, many studies have used heuristics to solve the NDP for a real-sized network in an acceptable computational time (*13*). To analyze the applicability of the proposed NDP on a real network, this study develops a heuristic algorithm based on a genetic algorithm (GA). With this heuristic algorithm, a case study is conducted on a container transportation network in the southeastern region of South Korea.

MODEL FORMULATION

In this study, a bilevel programming problem was established that represents the network planner's decision, which takes into account the responses of system users. On the basis of the objective function, the upper-level problem determines the decision variables, which represent actions for upgrading the level of the facilities. The "level of a facility" means a level that represents the capacity and mobility of the facility. It is defined by the attributes of the facility, such as capacity, speed, and free-flow time. The lower-level problem determines the traffic volume of each facility for the capacity that is decided at the upper level. Therefore, the problem was established as a mixed-integer, nonlinear, and bilevel programming problem.

This NDP is based on the following considerations:

- The network planner minimizes the total transportation cost of the freight network, and the users choose their mode and route to minimize their own transportation cost.
- The mode choice of transportation other than for targeted freight is neglected.
- The variability of the types of commodities is not considered, and the model focuses on container freight, which is favorable to rail transportation and is dominant and growing in use in interregional freight transportation in South Korea.
- The variability of the network users in terms of the value of travel time and their behaviors in choosing a transportation mode and route is not considered.
- The function of link travel time considers the characteristics of its mode. Rail and terminal links cannot accommodate volumes that exceed their capacity, and travel time is not sensitive to traffic volume when the volume is below capacity.
- The transfer time at a terminal link depends on the type and size of the handling equipment and the transfer track.

The upper-level problem was established as follows. The objective function of the upper-level problem consists of GHG emissions as well as travel time and transportation cost. Previous studies corresponding to strategic NDPs commonly used travel time, transportation cost, or a combination of the two as their objectives (*11, 12, 21–24*). This study introduced GHG emission cost to reduce GHG emissions. All of the cost terms in the objective function were calculated on a link basis. Through the use of a value of travel time and a value of GHG emissions, the objective function was set as a generalized cost. The constraints correspond to a budget limitation, maximum emissions on the freight network, and a limit on expansion or improvement at each facility. The volume of the link was determined by the lower-level problem.

$$\min_u Z = \mathrm{VOT} \cdot \sum_a t_{m_a}(u_a, x_a^*) \cdot x_a^* + \sum_a \mathrm{fare}_{m_a} \cdot x_a^* \cdot l_a + \mathrm{VOE} \cdot \sum_a e_{m_a}(u_a, x_a^*) \cdot x_a \cdot l_a^* \quad (1)$$

subject to

$$\sum_a g_a(u_a) \le B$$

$$\sum_a \left[l_a \cdot x_a^* \cdot e_{m_a}(u_a, x_a^*) \right] \cdot \gamma \le \max_E$$

$$0 \le u_a \le \max_u_a$$

x_a^* satisfies the lower-level problem

where

a = link,
m_a = mode of link a,
u_a = decision variable of link a,
x_a = freight volume on link a,
l_a = length of link a,
fare_m = freight rate of mode m,
VOT = value of time,
VOE = value of GHG emissions,
γ = yearly conversion factor,
$t_{m_a}(\cdot)$ = travel time function on link a of mode m,
$e_{m_a}(\cdot)$ = emission rate on link a of mode m,
$g_a(u_a)$ = expansion cost of link a,
B = allocated budget,
max_E = permitted maximum GHG emission, and
max_u_a = limit in improvement of link a.

In this formulation, the decision variable is an integer variable that corresponds to the facility's improvement action or the extent of upgrading the levels of the facility. For each mode, the levels of a facility were defined on the basis of the attributes that affect capacity. In this study, the levels of road and railroad links were defined by the number of lanes and railways, respectively. Terminal links were classified by the size and type of transfer track and loading equipment. An initial level was designated for each facility on the basis of its current attributes, and the decision variable of a facility was set as the number of levels of moving up from the initial level. With the definition of finite levels of a facility for each mode, the decision variable has a limitation for each facility (i.e., the limit in improvement is specified by the difference between the highest level and the initial level of the facility). With the decision variables set in this form, the NDP investigates improvement alternatives for all of the facilities in the network. In contrast, existing NDPs have dealt only with predefined and restricted alternatives.

Unlike previous studies, this formulation contains the mode-specific travel time function. To represent the congestion effect, a Bureau of Public Roads function was used as the link travel time function for the road link. For railroad and terminal links, an adjusted Davidson's function was used, which was revised on the basis of the function proposed in Akçelik's study (*25*). It can capture several characteristics: it cannot be used beyond the capacity, and link travel time is insensitive to link volume in an uncongested condition. As the capacity deficiency of a facility leads to rapid increase in travel time or restriction on usage, the NDP selects investment alternatives to reduce the total cost by reducing the link travel time or increasing the volume in low emission mode.

The emission rate, which represents the amount of GHG emissions produced by transporting one unit of volume (i.e., one 20-ft equivalent unit) one unit of distance (1 km), was also set as a mode-specific form. This rate was estimated on the basis of a fuel consumption rate, which depends on vehicle or equipment and type of fuel of each mode. For the road link, by setting the fuel consumption rate as a function of the link speed, the emission rate depends on the type of facility and the volume of the link. The emission rates for the railroad link and terminal facility were determined by the type of locomotive and the combination of transfer equipment, respectively. As a result, investing in a congested road, introducing electric locomotives, and changing transfer equipment can reduce total GHG emissions.

To represent the users' response to network improvement actions, users' travel choices for the given network condition were formulated as a lower-level problem. When the users do not own the network and do not use scheduled service, which are reasonable assumptions in South Korea, the user equilibrium assignment is acceptable. On a road network, other traffic directly affects the choice of the route of targeted freight. For these reasons, a multiclass user-equilibrium formulation based on Beckman's formulation was established as the lower-level problem.

$$\min_{\mathbf{v}} \sum_a \int_0^{v_a} c_a(u_a, w)\,dw = \sum_a \int_0^{v_a} \left[\text{VOT} \cdot t_{m_a}(u_a, w) + \text{fare}'_a \cdot l_a \cdot w\right] dw \tag{2}$$

subject to

$$\sum_k f^i_{k,rs} = q^i_{rs} \qquad \forall r, s, i \qquad f^i_{k,rs} \geq 0 \qquad \forall k, r, s, i$$

$$v^i_a = \sum_r \sum_s \sum_k f^i_{k,rs} \delta_{a,k,rs} \qquad v_a = \sum_i v^i_a \qquad \forall a$$

where

i = user class,
k = route,
r = origin,
s = destination,
$c_a(\cdot)$ = generalized cost on link a,
fare'_a = average unit fare of link a weighted by volumes of user classes,
v^i_a = traffic volume of class i on link a,
$f^i_{k,rs}$ = traffic volume of class i traveling on route k from r to s,
q^i_{rs} = travel demand of class i from r to s,
$\delta_{a,k,rs}$ = link incidence dummy variables,
v_a = traffic volume on link a converted to passenger car units,
$\mathbf{v}$ = vector of link traffic volume, and
w = dummy variable of integration.

In this study, the multimodal network was converted into a single, abstract-mode network, as mentioned in the study by Yamada et al., by using an expanded network representation with transfer links (*13*). This network representation technique rationally represents the intermodal transportation of the network and enables the use of conventional, node-labeled, shortest-path algorithms. In addition, it allows the cost on a transfer link to be represented in a functional form. By assigning transportation demand on this network, the lower-level problem simultaneously conducts the mode and route choices and produces equilibrium link traffic volumes for all modes.

SOLUTION ALGORITHM

Since it is a combinatorial optimization problem, a discrete NDP can be solved only through an enumeration technique. The computation time of an enumeration technique tends to increase exponentially as the search space becomes larger. For this reason, many discrete NDP studies have applied heuristic search algorithms to obtain effective solutions for real-size networks, as mentioned by Yamada et al. (*13*).

A bilevel discrete NDP could produce multiple local solutions because of the nonlinear function contained in the upper- and lower-level problems (*26*). For this reason, a GA, which is a population-based stochastic, global-search algorithm, was applied to solve the problem. Studies analyzing the applicability of various heuristics to NDPs have shown that a GA performs well (*27, 28*). In similar studies, the GA approach was used successfully (*13, 26, 29*). In recent years, many studies have been conducted to enhance the performance of GAs by combining a local search technique or other heuristics (*30–33*).

To solve this problem, a GA-based heuristic was established as shown in Figure 1. It stayed within the constraints by using a penalty function that had a large value for an individual that violated any constraint. The fitness function, which is a measure for evaluating each individual, was set up as Equation 3. Because the high penalty significantly lowered the degree of fitness, the GA could search for a feasible optimal solution.

$$F = \text{bigM} - Z - \eta \sum_j \phi_j \tag{3}$$

where

F = fitness of each individual,
bigM = arbitrary large positive number,
Z = objective function,
η = penalty parameter, and
ϕ_j = penalty function for constraint j.

To relieve the weakness of the general GAs, elitism was applied, and a feasibility check and the neighborhood improvement process were added. Elitism leaves the best individual in the current generation to the next generation. The feasibility check converts an infeasible solution into a feasible solution by adjusting part of its genes. The neighborhood improvement, which was introduced by Chainate et al. (*32*) and modified for the NDP in this study, investigates a better solution in its neighborhood on the basis of relevant information from the network. In this study, the volume-to-capacity ratios of links were used as information for the neighborhood improvement. By activating the feasibility check and neighborhood improvement step stochastically, the GA tried to balance between performing a fine search and a global search.

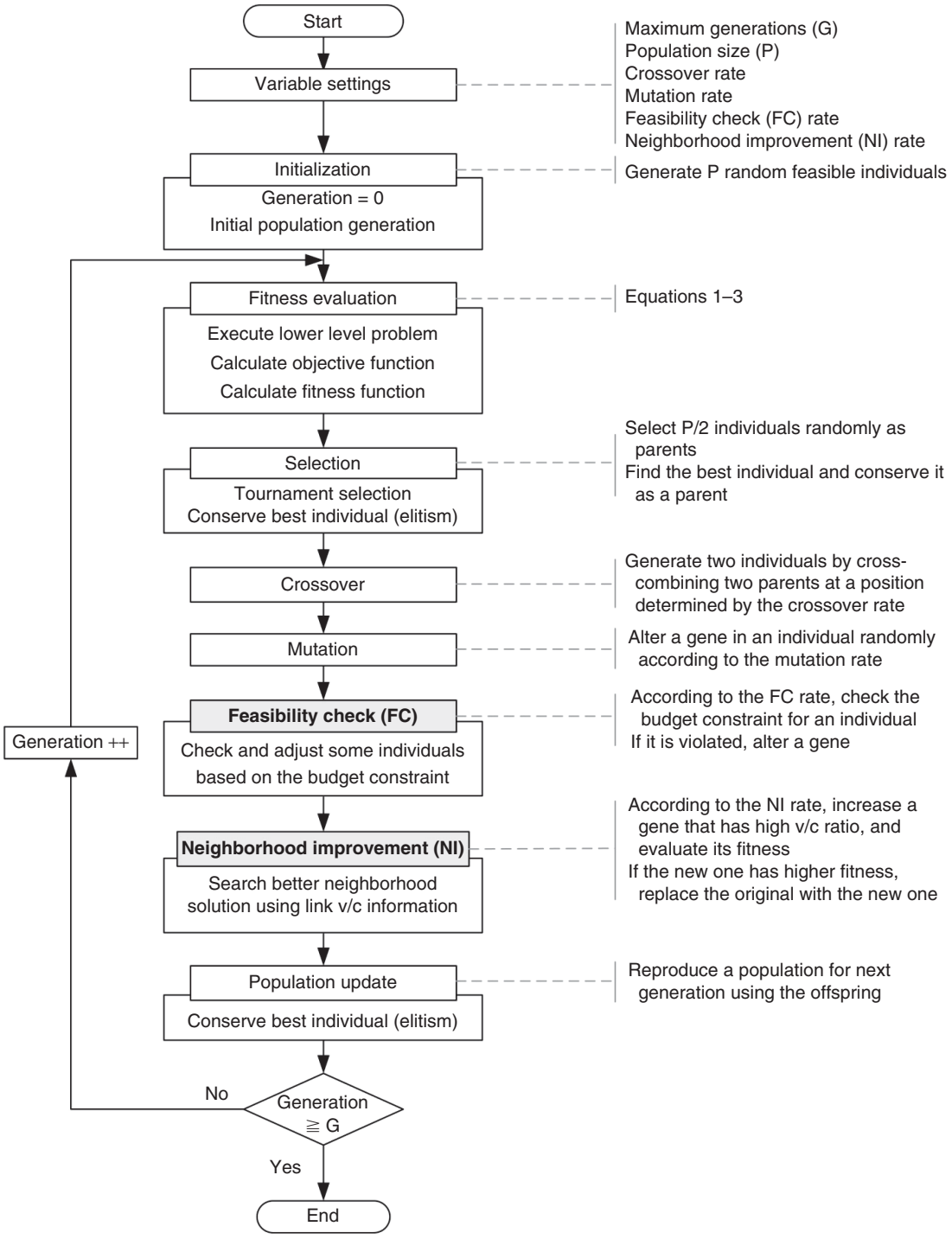


FIGURE 1 Process of GA used in study (v/c = volume to capacity).

To test the performance, the GA solution was compared with the optimal solution through an enumeration technique for a hypothetical small network consisting of 28 modal links. As the search space became larger because of the increase of levels of facility or rise in budget, the computation time of the enumeration technique increased exponentially. In contrast, the computation time of the GA depended on the number of generations and the population size. For this example, the computation was completed within 180 s in the case of 1,000 generations and a population size of 100. Through repetitive use, the GA found an acceptable, effective solution within a relatively short time, although its performance declined slightly with the increase in search space.

APPLICATION AND RESULTS

The NDP formulated in this study was applied to the interregional container transportation network from the Port of Pusan, South Korea, to major destinations in the southeastern part of the country. The Port of Pusan is the largest container import–export port in South Korea; it handled about 11,980,000 twenty-ft equivalent units in 2009. A container transportation demand matrix from the Port of Pusan to eight major destinations was used as input on the basis of import container data provided by the Korea Transport Database. Most of the container freight was moved toward the Seoul, South Korea, metropolitan area and the southeast region.

For the application of the NDP addressed in this study, a simplified network was constructed that consisted of 26 express road links, 24 railroad links, and eight transfer terminals, as shown in Figure 2. Two inland container terminals and six railroad container yards were considered as terminals. Among the railroad lines, the Gyeongju and the Pusan New Port lines, which connect Pusan and the Seoul metropolitan area, have double-electrified tracks, while other lines have a single track. Although operational electrified tracks are available, electric locomotives are seldom operated for freight transportation.

Table 1 shows the predetermined facility levels for the application. Although freight rail cars are pulled by diesel locomotives, the operation of electric locomotives was considered to analyze the effect of changing the energy source. Introducing an electric locomotive is an alternative that can raise the speed of rail service with a comparatively low investment cost. The addition of transfer equipment and transfer tracks at the terminal was considered to reduce transfer time and increase capacity.

Table 2 shows the parameters and input values for an empirical application. Most of the values were adopted from guidelines and

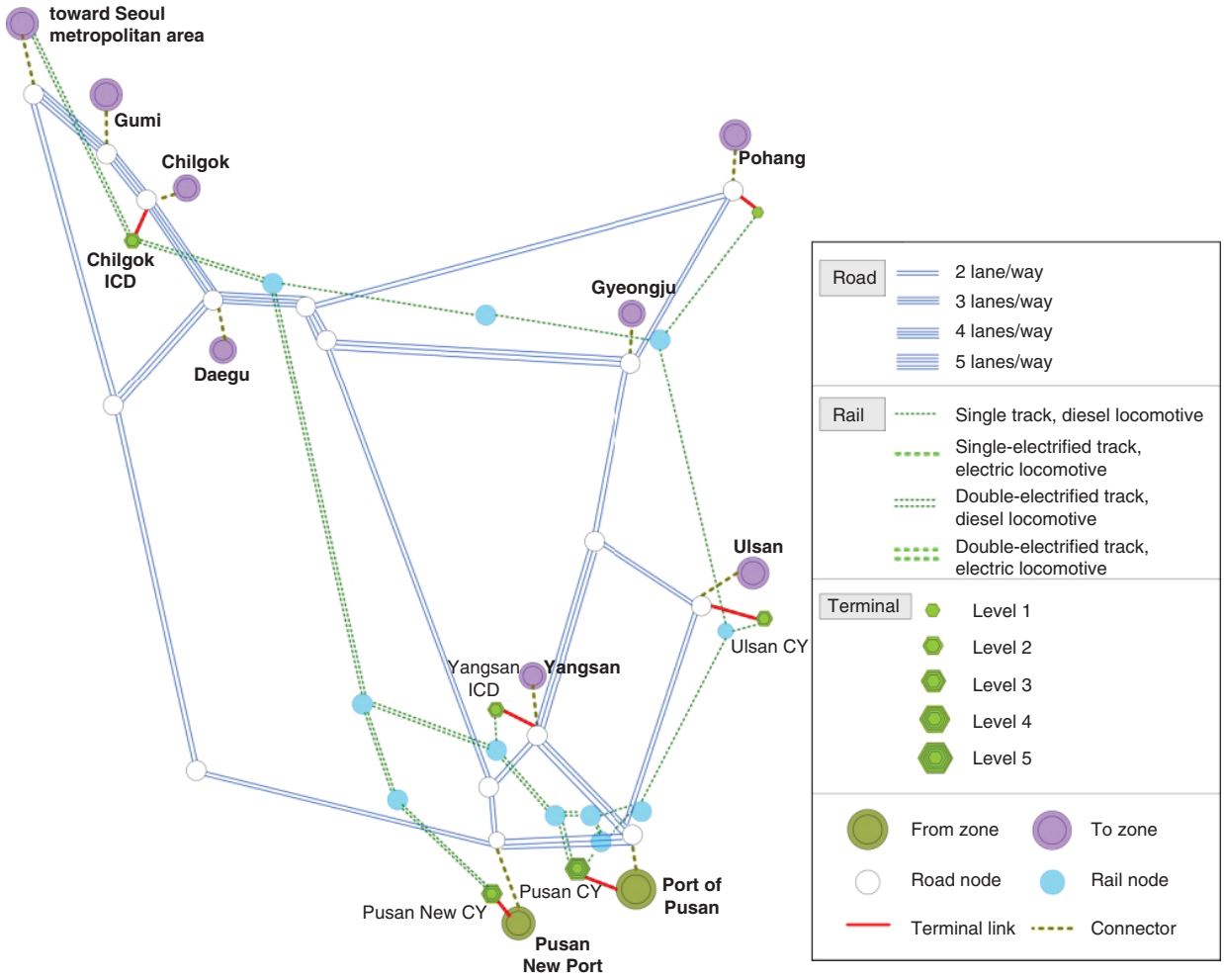


FIGURE 2 Container transportation network for application (CY = container yard; ICD = inland container depot).

TABLE 1 Design Levels and Attributes

Mode	Design Level	Facility Type	Capacity (PCU, train, TEU/day)	Free Speed (km/h)	Uncongested Transfer Time (h)
Road	1	2 lanes	25,000	80	—
	2	3 lanes	60,000	90	—
	3	4 lanes	80,000	100	—
	4	5 lanes	100,000	100	—
Rail	1	Single track, diesel locomotive	30	50	—
	2	Single-electrified track, electric locomotive	50	90	—
	3	Double-electrified track, diesel locomotive	170	70	—
	4	Double-electrified track, electric locomotive	170	90	—
Terminal	1	Transfer track 25 m × 2, RS × 1	300	—	4.3
	2	Transfer track 25 m × 3, TC × 1 + RS × 1	500	—	2.5
	3	Transfer track 25 m × 4, TC × 2 + RS × 1	800	—	2.0
	4	Transfer track 30 m × 6, TC × 3 + RS × 2	1,300	—	1.7
	5	Transfer track 30 m × 8, TC × 4 + RS × 2	1,800	—	1.7

NOTE: PCU = passenger car units, RS = reach stacker, TC = transfer crane, TEU = 20-ft equivalent unit. — = not applicable.

previous studies that are commonly referred to in South Korea. Among inputs other than those in Table 2, investment cost for facility expansion was based on a relevant guide and previous investment projects in that area. In addition, standard rates for container transportation in South Korea were applied for the road and rail freight rates.

By using the freight demand matrix, facility levels, and other input data, the NDP developed in this study was applied to the study network. Under the 5 trillion won budget constraint (1,000 won = US$0.92 on November 9, 2012) and the 174.6 kiloton emission constraint (18% reduction from the current condition), the solution of the NDP is shown in Figure 3. To satisfy the emissions reduction target, improvement actions on railroads and terminals were selected, along with a few link expansion alternatives on congested roads. For corridors that have high-level railroad systems such as those leading to the Gumi (South Korea) and Seoul metropolitan areas, investment was concentrated on the rail and transfer facilities. In contrast, expansions of congested road links were selected for corridors with low-level rail systems, such as Pohang, Ulsan, and Gyeongju, South Korea.

The difference between container freight volumes on the current and the solution networks can be seen in Figure 4. After the NDP solution was applied, volumes on the railroad lines toward the Seoul metropolitan area increased, while traffic volumes on roads in the same direction decreased. Consequently, modal shift of the transportation demand toward Seoul and Gumi would reduce the system cost and mitigate congestion on road links.

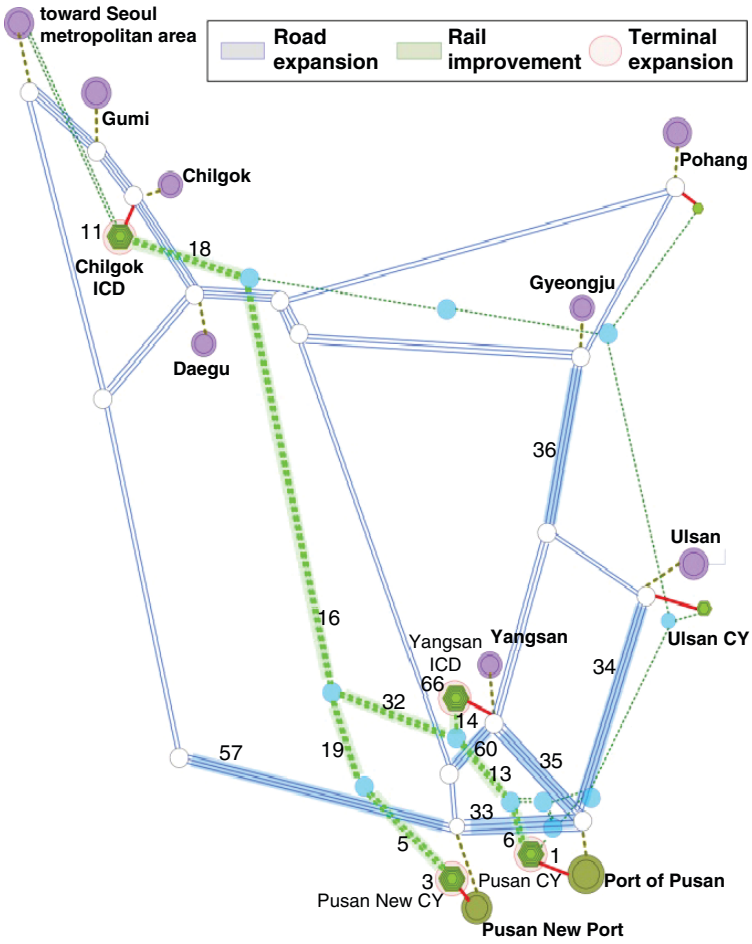
To verify the NDP established in this study, solutions for the case with and without the emission constraint and the emission cost under the same budget limit were investigated, as shown in Figure 5. Remarkable differences were found for the railroad and terminal links. The solution of the NDP without consideration of GHG emissions showed that investment would be concentrated in the roads. Including the emission cost in the objective function and the emission constraint increased the investment in rail and terminal facilities. Because of the differences in emission rates, improving the railroad and terminal facilities and attracting more freight to railroads reduced emissions. In addition, restricted and selective investment on congested roads decreased travel time and emissions.

Table 3 shows the differences in total system cost and the cost elements for the four cases in Figure 5. Compared with the current status, all NDP solutions reduced total system cost, but they showed differences in each cost element. First, investments not considering emissions reduction increased the emissions despite the capacity improvements. In addition, the emissions constraint allowed meeting

TABLE 2 Parameters and Input Values

Parameter	Mode	Type	Input Value	Source or Background
Parameters of link travel time function	Road	BPR function	$\alpha = 0.611$ $\beta = 2.772$	Recommended by KTDB
	Rail	Davidson's function	$J = 0.01$	
	Terminal	Davidson's function	$J = 0.10$	
Emission rate (CO_2e kg/veh-km for road and rail, CO_2e kg/TEU for terminal)	Road	A function of link travel speed		Estimated by fuel-consumption rates and emission factors from the IPCC Guidelines (*34*)
	Rail	Diesel locomotive	0.3625	Based on statistics, emission factor (*35–37*) and Korea Power Exchange's announcement (*38*)
		Electric locomotive	0.1958	
	Terminal	Transfer crane	2,972	Calculated using energy consumption rates from previous studies (*39*, *40*)
		Reach stacker	1.857	
Value of travel time (won/TEU-h)	All		7,936	Study in Korea (*41*)
Value of GHG emission (won/ton CO_2)	All		150,000	Guideline in Korea (*42*)

NOTE: 1,000 won = US$0.92 on November 9, 2012. BPR = Bureau of Public Roads, CO_2 = carbon dioxide, CO_2e = carbon dioxide equivalent, veh-km = vehicle kilometers, IPCC = Intergovernmental Panel on Climate Change, KTDB = Korea Transport Database.



Mode	Link index	Alternative description
Road	60	2 lanes expansion
	33, 34, 35, 36, 57	1 lane expansion
Rail	14	Rail electrification and electric locomotive introduction
	5, 6, 13, 16, 18, 19, 32	Electric locomotive introduction
Terminal	11, 66	3 level expansion
	3	2 level expansion
	1	1 level expansion

FIGURE 3 NDP solution (budget constraint = 5 trillion won, emission constraint = 174.6 kiloton).

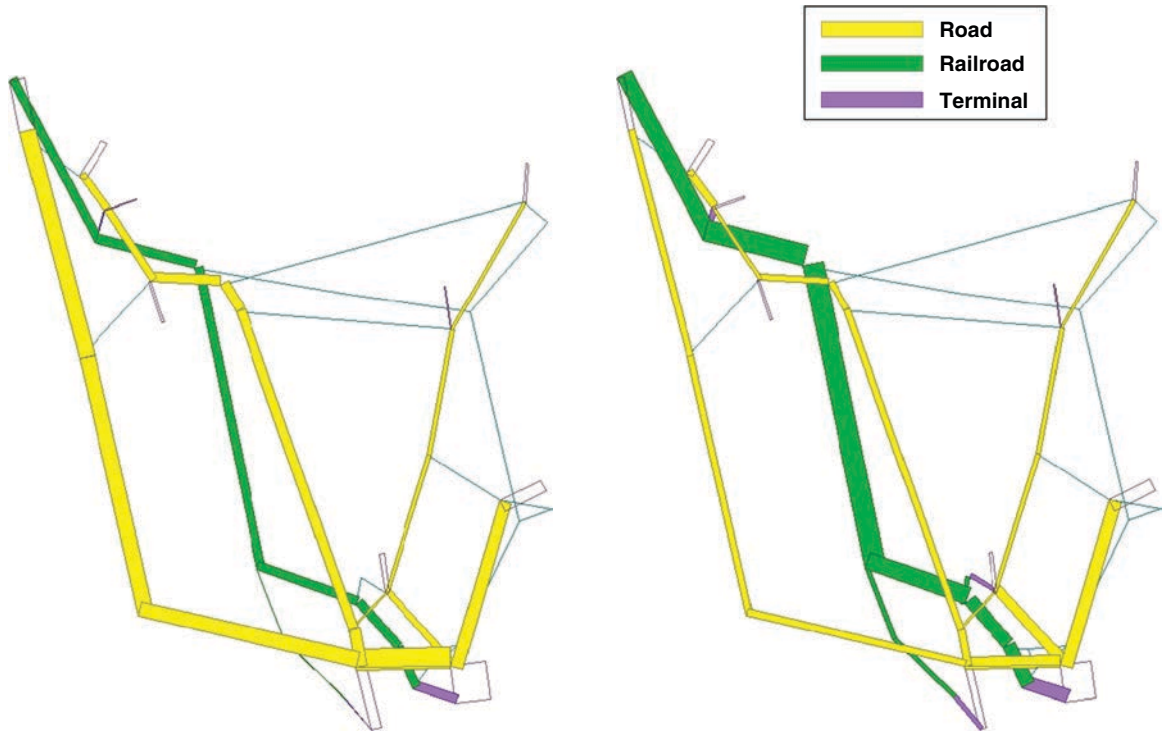


FIGURE 4 Change of container volume by NDP solution: (*a*) volume on current network and (*b*) volume on solution network. (Thickness of a line is proportional to its volume.)

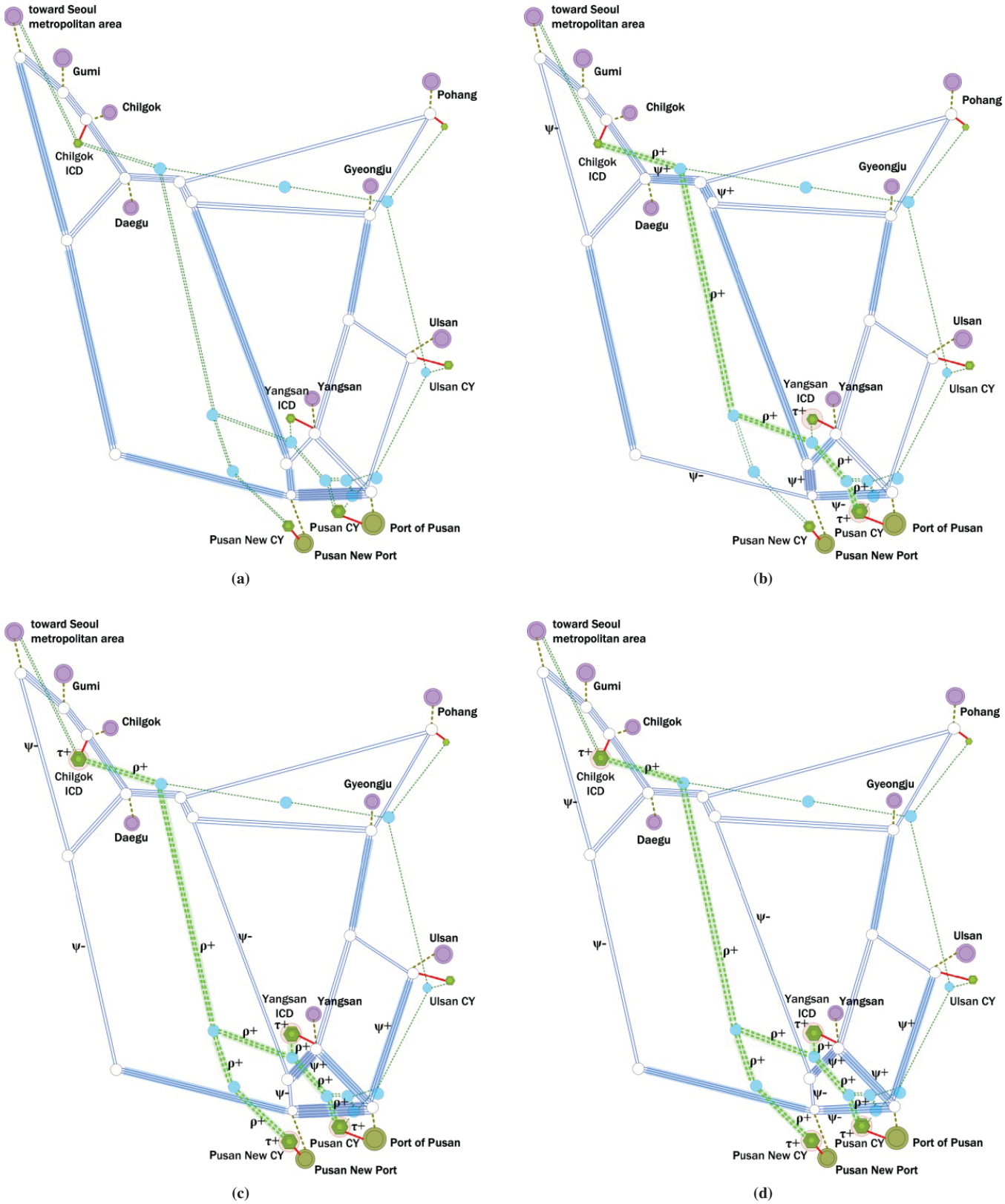


FIGURE 5 NDP solutions with and without emission cost and constraint under same budget: (*a*) not considering emission reduction, (*b*) including emission cost and not considering emission constraint, (*c*) including emission constraint and not considering emission cost, and (*d*) including emission cost and constraint. [$\rho+$ (or $\rho-$) = increase (or decrease) in rail investment of indicated link compared with that in (*a*), $\psi+$ (or $\psi-$) = increase (or decrease) in road investment of indicated link compared with that in (*a*), and $\tau+$ (or $\tau-$) = increase (or decrease) in terminal investment of indicated link compared with that in Solution *a*.]

TABLE 3 Comparisons of Solutions With and Without Emissions Constraint and Emissions Cost

		Without Emissions Constraint		With Emissions Constraint (174.6 kiloton, 18% reduction)	
Cost (in billion won)	Do Nothing	Without Emission Cost in Objective Function	With Emission Cost (TEC)	Without Emission Cost in Objective Function	This Model (TEC)
Total system cost (TEC)	306.9	302.7	294.2	295.6	294.8
Travel time cost (T)	41.2	25.7	42.9	44.9	46.5
Emission cost (E)	31.9	35.4	26.7	26.2	25.4
Fare (C)	233.7	241.6	224.7	224.5	223.0
Investment	—	4,805.3	4,916.5	4,969.9	4,950.4
GHG emissions (kton)	212.9	235.8	177.8	174.5	169.1

the emissions reduction target, and the emission cost term reduced the total system cost of the network. In other words, without an emissions constraint, the network planner would make investment decisions that fail to meet the emissions reduction target. Similarly, excluding emission cost leads to an excessive social cost. Compared with the case without an emission cost term, the solution of this model caused travel time and transportation cost (T + C) to increase by 70 million won, but it reduced the total cost by as much as 740 million won.

In addition, changes of network planning circumstances, such as budget amount, the value or social unit cost of GHG emissions, and strength of the emission reduction target, were investigated. The results indicate that as the pressure of emissions reduction grows because of a higher reduction target or value of emissions, investment was more concentrated in railroads and terminal facilities. In addition, comparison of various values of maximum emissions showed that the total system cost tends to increase as the reduction target strengthens. With a low budget, low-cost emissions reduction alternatives, such as introducing electric locomotives and improving terminals, were selected. When the budget was high, road expansion and high-cost railroad enhancement alternatives, such as electrifying the railroad and adding more tracks, were selected.

CONCLUSIONS

This study developed a multimodal container freight NDP to reduce GHG emissions below a target. By including emission cost in the objective function and the maximum emission constraint in the upper-level problem, the NDP provided an optimal combination of investment alternatives to minimize the total system cost and to meet the emission reduction target. On the basis of a GA established for this study, a case study of a container transportation network from the Port of Pusan was investigated.

The results showed that this NDP could provide an investment decision to achieve the policy objectives. It was found that containing the emission cost and the emission constraint drove the solution to reduce the total system cost and to achieve the required emission reduction target, respectively. In addition, the results indicated that investment should focus on railroad and terminal facilities to induce a modal shift, whereas investment in congested road links could be a better option for corridors in which the level of the railroad system was low and short-distance freight demands were dominant.

The results have significant implications. First, an NDP that does not consider the emission cost tends to drive investment toward roads. This distorts usage of the transportation network in view of the social cost of GHG emissions. Therefore, the NDP developed in this paper can contribute to the correction of that distortion. Second, an NDP without an emission constraint is not likely to meet the emissions reduction target. When emissions reduction is required by international agreement, achieving the target becomes obligatory. This implies that an emission constraint should be included in NDPs under that requirement. Third, the present NDP considered investment alternatives in the transfer terminals and applied a travel time function that depends on the freight volume and capacity. These features resulted in the solution's being concentrated on improving the terminals as well as the railroad links. This suggests that the planner should save transfer time along with railroad line-haul time to lead a modal shift.

This NDP has limitations and issues that need further study. It applied input values that represent average container freight. The NDP could be enhanced to consider different shipment sizes and vehicle classes and their compositions in container transportation. The model could be expanded to accommodate various commodities, even though that requires careful investigation. In addition, the travel time function for a transfer facility has a weak foundation, so revision of the function and estimation of the parameters on the basis of reliable data to strengthen its validity are necessary.

This study can be used in the following ways. It can help in making decisions with regard to the investment planning process under the GHG emissions reduction policy. It can provide the basis for the argument that investments in energy-efficient modes should be augmented, as stated in several policy reports (*4, 43*). It can provide the target modal split for optimizing the system's objectives and for meeting the emission mitigation goal. In addition, the NDP can help in coping with changes in planning circumstances, such as budget fluctuations, emission values, and target emissions reduction levels. When the pressure to reduce emissions increases, support planning that lowers emission rates should be enhanced.

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